

# **PARALLEL DATABASE SYSTEMS**

## **UNIT V**

## DEFINITION

- Parallel DBMS is a Database Management System that runs through multiple processors and disks.
- They combine two or more processors also disk storage that helps make operations and executions easier and faster.
- They are designed to execute concurrent operations.
- They exist, happen, or done at the same time even if the data processed are not from one source or one processing unit.

## NEED

- Multiple resources like CPUs and Disks are used in parallel.
- The operations are performed simultaneously, as opposed to serial processing.
- A parallel server can allow access to a single database by users on multiple machines.
- It also performs many parallelization operations like data loading, query processing, building indexes, and evaluating queries.

# ARCHITECTURE

The main architecture for parallel DBMS is:

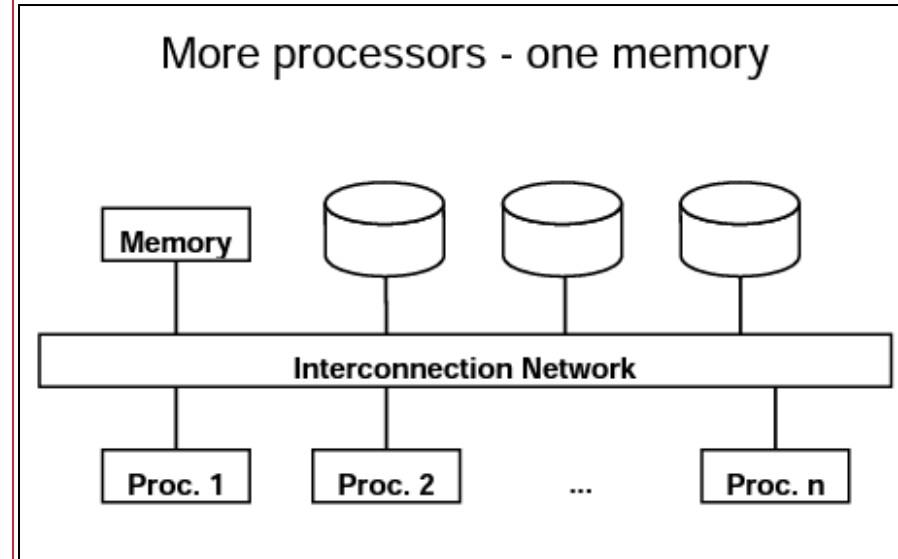
1. Shared Memory System
2. Shared Disk System
3. Shared Nothing System
4. Hybrid architectures

# 1. SHARED MEMORY SYSTEM

- A Shared Memory System is an architecture of Database Management System, where every computer processor is able to access and process data from multiple memory modules or unit through intercommunication channel.
- This architecture is also commonly known as SMP or Symmetric Multi-processing.
- A Shared Memory System contains large amounts of cache memories at each processor, so referencing of the shared memory is avoided.

## Characteristics

*any CPU has access to any memory module or disk unit*



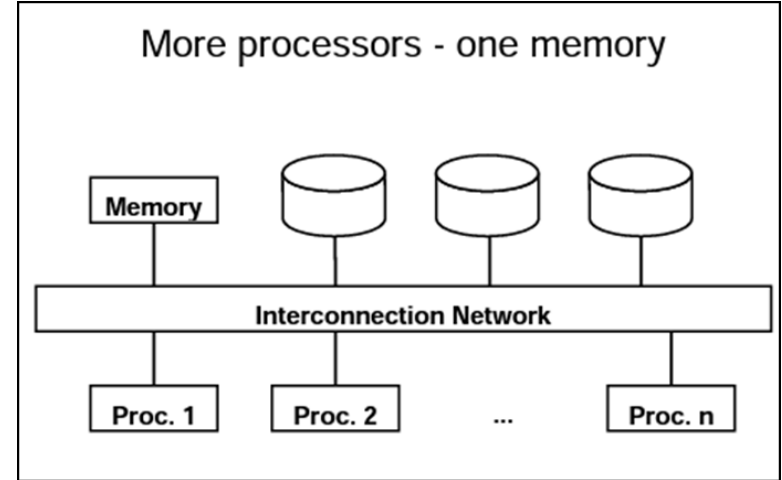
# 1. SHARED MEMORY SYSTEM

## *Advantages of Shared Memory:*

- Data is easily accessed from various processors.
- A single processor can send messages to other processors efficiently.

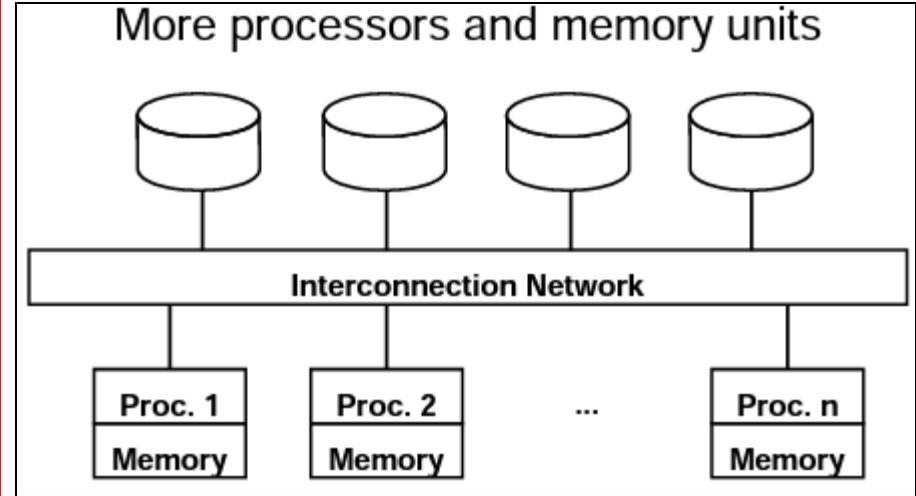
## *Disadvantage of Shared Memory:*

- Waiting time for every single processor increases when all of them are uses.
- Bandwidth is also a problem.



## 2. SHARED DISK SYSTEM

- A Shared Disk System is an architecture of Database Management System where every computer processors can access multiple disk through intercommunication network.
- It can also access and utilize every local memory.
- Each of the processors have own memory system, so the shared data are more efficient.



## 2. SHARED DISK SYSTEM

### *Advantages of Shared Disk System:*

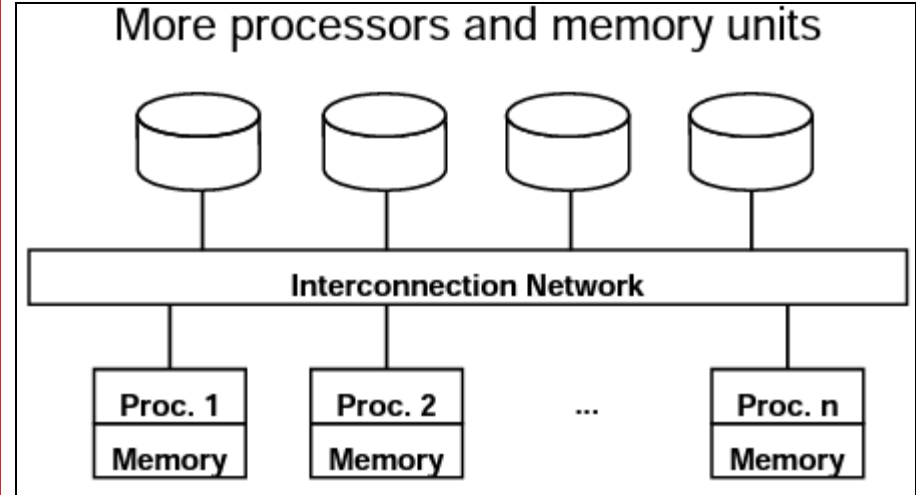
- The fault tolerance can be achieved using this system.

### *Disadvantages of Shared Disk System:*

- The addition of processors can slow down existing processors.
- Shared Disk System have a limited scalability meaning it is sometimes fixed.

### **Characteristics**

*any CPU has access to any disk unit but exclusive access to its main memory*



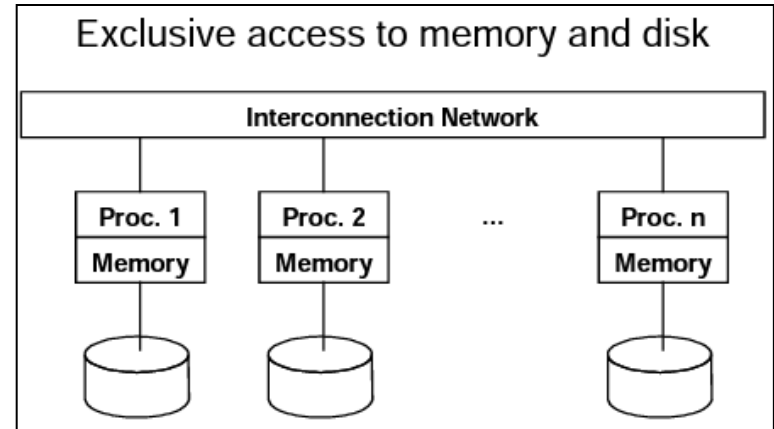


### 3. SHARED NOTHING SYSTEM

- A Shared Nothing System is an architecture of Database Management System where every processor has their own disk and memory for the objective of efficient workflows.
- The processors can communicate with other processors using intercommunication network.
- Each of the processors act like servers to store data on the disk. So there can be an efficient and effective workflows.

#### Characteristics

*any CPU has only exclusive access to its main memory and disk*



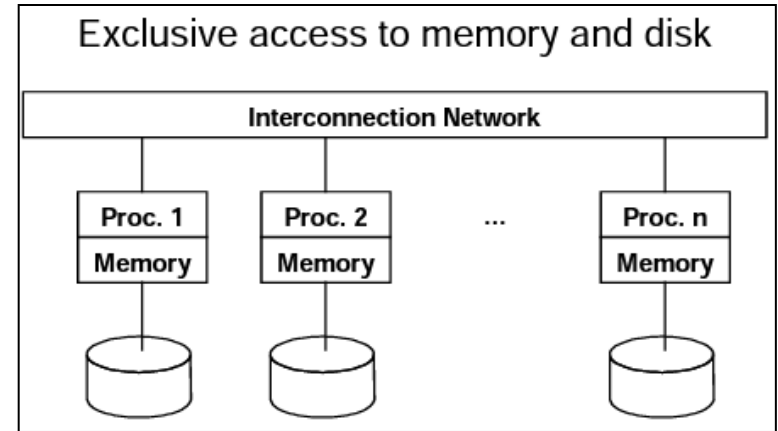
### 3. SHARED NOTHING SYSTEM

#### Advantage of Shared Nothing System

- This system has more scalability.
- Number of processors and disk can be connected as per the requirement in share nothing disk system.

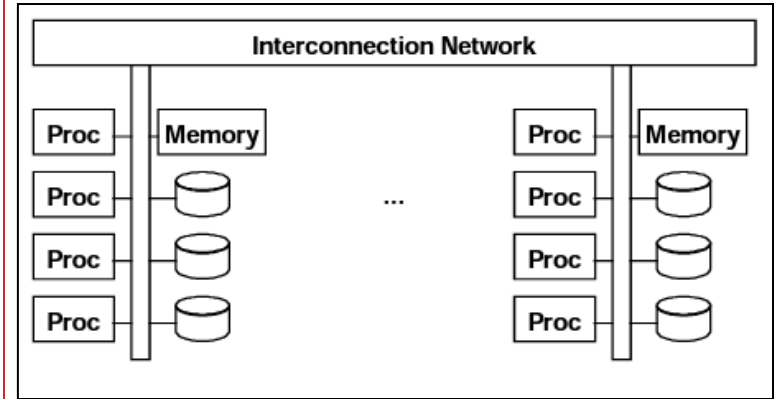
#### Disadvantage of Shared Nothing system

- Must require the partitioning of data.
- Cost that need for this system is higher.



### 3. HYBRID ARCHITECTURES SYSTEM

- combine the advantages of different architectures
- use different processing elements in the system
- The system is basically a shared-nothing architecture where each node can be a multicomputer of any architecture.



# ADVANTAGES OF PARALLEL DBMS

- **Speed**

The first advantage of parallel DBMS is speed. The servers from parallel DBMS are able to break up user database request into parts and it dispatches each of the parts to separate computers. They work on these “work parts” simultaneously and they merge the results, passing them back to the user. This speeds up most of the data requests, allowing faster access to very large databases.

- **Reliability**

The second advantage is reliability. A parallel database when properly configured, can continue to work despite the failure of any computer in the cluster. The database server can sense that a specific computer is not responding, and it can reroute its work to the remaining computers.

- **Capacity**

The third advantage is capacity. As more and more users request access to the database, the computer administrators add more computers to the parallel server, boosting its overall capacity to the max, databases are more likely to slow down and have slower system. A parallel database, for example, allows a large online retailer to have thousands of requests of users accessing information at the same time. This level of processing performance is not possible with single server systems.

## **DISADVANTAGE OF PARALLEL DBMS :**

- **Cost**

The first disadvantage of parallel DBMS is cost. As the need of quicker processing and efficient searches increase, there need to be more and more disks and processors that simultaneously work together to achieve the best and quickest results. To do that, Parallel DBMS needs lots of processors and disks in the first place. At the end, it is never cheap to implement parallel DBMSs.

- **Resources**

The second disadvantage of parallel DBMS is the huge amount of resources. It is not and will never be easy to keep up with modern, cheap, and efficient resources. Implementing DBMS requires the users or the company to have renewal of resources, changing or maintaining resources, or even replacement of resources.

- **Difficulty of Managing Systems**

The third disadvantage of parallel DBMS is the difficulty in systems managing. When having lots of systems, lots of resources, and lots of systems running it is never easy to manage them. When there needs to be a software update, a replacement, or maintenance that all of the system needs to do, it will be time consuming and resource consuming.

# DIFFERENCE BETWEEN PARALLEL AND DISTRIBUTED DATABASES

| Sl No | Parallel Database                                                                                                                                                                        | Distributed Database                                                                                                                                                                               |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1     | In parallel databases, processes are tightly coupled and constitutes a single database system i.e., the parallel database is a centralized database and data reside in a single location | In distributed databases, the sites are loosely coupled and share no physical components i.e., distributed database is our geographically departed, and data are distributed at several locations. |
| 2     | In parallel databases, query processing and transaction is complicated.                                                                                                                  | In distributed databases, query processing and transaction is more complicated.                                                                                                                    |
| 3     | In parallel databases, it's not applicable.                                                                                                                                              | In distributed databases, a local and global transaction can be transformed into distributed database systems                                                                                      |

# DIFFERENCE BETWEEN PARALLEL AND DISTRIBUTED DATABASES

| SI No | Parallel Database                                                                                                | Distributed Database                                                                                                                     |
|-------|------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| 4     | In parallel databases, the data is partitioned among various disks so that it can be retrieved faster.           | In distributed databases, each site preserve a local database system for faster processing due to the slow interconnection between sites |
| 5     | In parallel databases, there are 3 types of architecture: shared memory, shared disk, and shared shared-nothing. | Distributed databases are generally a kind of shared-nothing architecture                                                                |
| 6     | In parallel databases, query optimization is more complicated.                                                   | In distributed databases, query Optimisation techniques may be different at different sites and are easy to maintain                     |

# DIFFERENCE BETWEEN PARALLEL AND DISTRIBUTED DATABASES

| Sl No | Parallel Database                                                                        | Distributed Database                                                                                      |
|-------|------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| 7     | In parallel databases, data is generally not copied.                                     | In distributed databases, data is replicated at any number of sites to improve the performance of systems |
| 8     | Parallel databases are generally homogeneous in nature                                   | Distributed databases may be homogeneous or heterogeneous in nature.                                      |
| 9     | Skew is the major issue with the increasing degree of parallelism in parallel databases. | Blocking due to site failure and transparency are the major issues in distributed databases.              |



## PARALLELISM IN QUERY IN DBMS

- Parallelism in a query allows us to parallel execution of multiple queries by decomposing them into the parts that work in parallel.
- This can be achieved by shared-nothing architecture. Parallelism is also used in fastening the process of a query execution as more and more resources like processors and disks are provided.

# METHODS

- I/O parallelism
- Intra-query parallelism
- Inter-query parallelism
- Intra-operation parallelism
- Inter-operation parallelism

# I/O PARALLELISM

- It is a form of parallelism in which the relations are partitioned on multiple disks a motive to reduce the retrieval time of relations from the disk.
- Within, the data inputted is partitioned and then processing is done in parallel with each partition. The results are merged after processing all the partitioned data. It is also known as data-partitioning.
- Hash partitioning has the advantage that it provides an even distribution of data across the disks and it is also best suited for those point queries that are based on the partitioning attribute.
- It is to be noted that partitioning is useful for the sequential scans of the entire table placed on 'n' number of disks and the time taken to scan the relationship is approximately  $1/n$  of the time required to scan the table on a single disk system.

# I/O PARALLELISM

We have four types of partitioning in I/O parallelism:

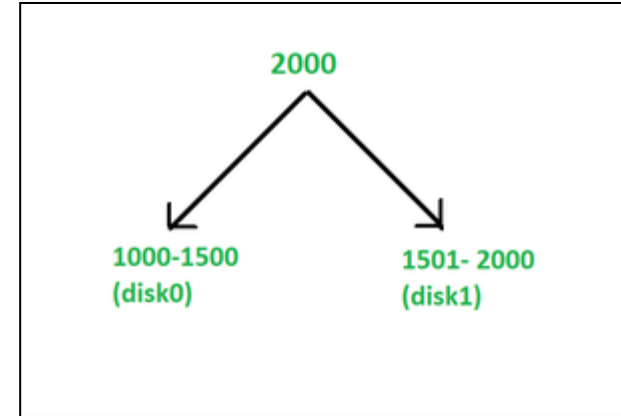
- Hash partitioning
- Range partitioning
- Round-robin partitioning
- Schema partitioning

# HASH PARTITIONING

- A Hash Function is a fast, mathematical function.
- Each row of the original relationship is hashed on partitioning attributes.
- For example, let's assume that there are 4 disks disk1, disk2, disk3, and disk4 through which the data is to be partitioned. Now if the Function returns 3, then the row is placed on disk3.

# RANGE PARTITIONING

- In range partitioning, it issues continuous attribute value ranges to each disk.
- For example, we have 3 disks numbered 0, 1, and 2 in range partitioning, and may assign relation with a value that is less than 5 to disk0, values between 5-40 to disk1, and values that are greater than 40 to disk2.
- It has some advantages, like it involves placing shuffles containing attribute values that fall within a certain range on the disk.

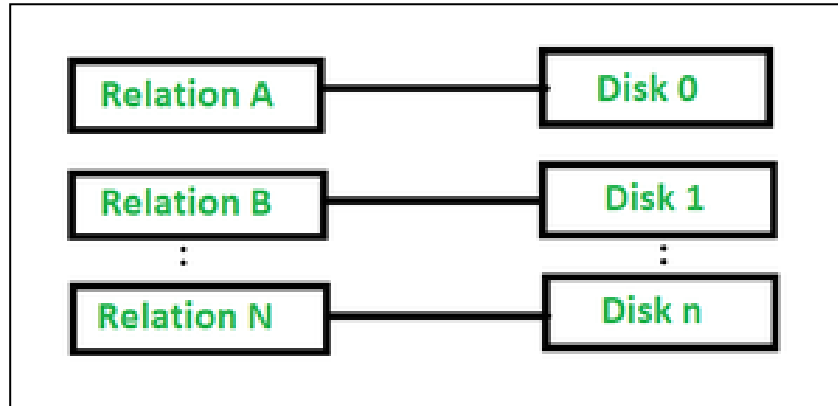


## ROUND-ROBIN PARTITIONING

- In Round Robin partitioning, the relations are studied in any order. The  $i$ th tuple is sent to the disk number  $(i \% n)$ .
- So, disks take turns receiving new rows of data.
- This technique ensures the even distribution of tuples across disks and is ideally suitable for applications that wish to read the entire relation sequentially for each query.

# SCHEMA PARTITIONING

In schema partitioning, different tables within a database are placed on different disks.





## 2. INTRA-QUERY PARALLELISM :

- Intra-query parallelism refers to the execution of a single query in a parallel process on different CPUs using a shared-nothing paralleling architecture technique.
- This uses two types of approaches:
  - ✓ First approach –  
In this approach, each CPU can execute the duplicate task against some data portion.
  - ✓ Second approach –  
In this approach, the task can be divided into different sectors with each CPU executing a distinct subtask.

### 3. INTER-QUERY PARALLELISM

- ✓ In Inter-query parallelism, there is an execution of multiple transactions by each CPU. It is called parallel transaction processing.
- ✓ DBMS uses transaction dispatching to carry inter query parallelism.
- ✓ We can also use some different methods, like efficient lock management.
- ✓ In this method, each query is run sequentially, which leads to slowing down the running of long queries.
- ✓ In such cases, DBMS must understand the locks held by different transactions running on different processes.
- ✓ Inter query parallelism on shared disk architecture performs best when transactions that execute in parallel do not accept the same data.
- ✓ Also, it is the easiest form of parallelism in DBMS, and there is an increased transaction throughput.

## 4. INTRA-OPERATION PARALLELISM :

- Intra-operation parallelism is a sort of parallelism in which we parallelize the execution of each individual operation of a task like sorting, joins, projections, and so on.
- The level of parallelism is very high in intra-operation parallelism. This type of parallelism is natural in database systems.

Let's take an SQL query example:

```
SELECT * FROM Vehicles ORDER BY Model_Number;
```

In the above query, the relational operation is sorting and since a relation can have a large number of records in it, the operation can be performed on different subsets of the relation in multiple processors, which reduces the time required to sort the data.

## 5. INTER-OPERATION PARALLELISM :

- When different operations in a query expression are executed in parallel, then it is called inter-operation parallelism.
- They are of two types –
  - ✓ ***Pipelined parallelism*** –  
In pipeline parallelism, the output row of one operation is consumed by the second operation even before the first operation has produced the entire set of rows in its output. Also, it is possible to run these two operations simultaneously on different CPUs, so that one operation consumes tuples in parallel with another operation, reducing them. It is useful for the small number of CPUs and avoids writing of intermediate results to disk.
  - ✓ ***Independent parallelism*** –  
In this parallelism, the operations in query expressions that are not dependent on each other can be executed in parallel. This parallelism is very useful in the case of the lower degree of parallelism.

Thank You

Thank You